

LLMs for classification and regression

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ChatGPT outperforms crowd workers for text-annotation tasks

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Many NLP applications require manual text annotations for a variety of tasks, notably to train classifiers or evaluate the performance of unsupervised models. Depending on the size and degree of complexity, the tasks may be conducted by crowd workers on platforms such as MTurk as well as trained annotators, such as research assistants. Using four samples of tweets and news articles ($n = 6,183$), we show that ChatGPT outperforms crowd workers for several annotation tasks, including relevance, stance, topics, and frame detection. Across the four datasets, the zero-shot accuracy of ChatGPT exceeds that of crowd workers by about 25 percentage points on average, while ChatGPT's intercoder agreement exceeds that of both crowd workers and trained annotators for all tasks. Moreover, the per-annotation cost of ChatGPT is less than \$0.003—about thirty times cheaper than MTurk. These results demonstrate the potential of large language models to drastically increase the efficiency of text classification.

ChatGPT | text classification | large language models | human annotations | text as data



Finetuning LLMs for Human Behavior Prediction in Social Science Experiments

Akaash Kolluri, Shengguang Wu, Joon Sung Park, Michael S. Bernstein

Abstract

Large language models (LLMs) offer a powerful opportunity to simulate the results of social science experiments. In this work, we demonstrate that finetuning LLMs directly on individual-level responses from past experiments meaningfully improves the accuracy of such simulations. We construct SocSci210 via an automatic pipeline, a dataset comprising 2.9 million responses from 400,491 participants in 210 open-source social science experiments. Through finetuning, we achieve multiple levels of generalization. In completely unseen studies, our strongest model, Socrates-Qwen-14B, produces predictions that are 36% more aligned with distributions of human responses to diverse outcome questions under varying conditions relative to its base model (Qwen2.5-14B), outperforming GPT-4o by 15%. By finetuning on a subset of conditions in a study, generalization to new unseen conditions is particularly robust, improving by 71%. Since SocSci210 contains rich demographic information, we reduce demographic parity difference, a measure of bias, by 10.6% through finetuning. Because social sciences routinely generate rich, topic-specific datasets, our findings indicate that finetuning on such data could enable more accurate simulations for experimental hypothesis screening. We release our data, models and finetuning code.

[PDF](#)[Cite](#)[Search](#)[Checklist](#)[Fix data](#)

A foundation model to predict and capture human cognition

[Marcel Binz](#) , [Elif Akata](#), [Matthias Bethge](#), [Franziska Brändle](#), [Fred Callaway](#), [Julian Coda-Forno](#), [Peter Dayan](#), [Can Demircan](#), [Maria K. Eckstein](#), [Noémi Éltető](#), [Thomas L. Griffiths](#), [Susanne Haridi](#), [Akshay K. Jagadish](#), [Li Ji-An](#), [Alexander Kipnis](#), [Sreejan Kumar](#), [Tobias Ludwig](#), [Marvin Mathony](#), [Marcelo Mattar](#), [Alireza Modirshanechi](#), [Surabhi S. Nath](#), [Joshua C. Peterson](#), [Milena Rmus](#), [Evan M. Russek](#), ... [Eric Schulz](#) [+ Show authors](#)

[Nature](#) **644**, 1002–1009 (2025) | [Cite this article](#)

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Abstract

Establishing a unified theory of cognition has been an important goal in psychology^{1,2}. A first step towards such a theory is to create a computational model that can predict human behaviour in a wide range of settings. Here we introduce Centaur, a computational model that can predict and simulate human behaviour in any experiment expressible in natural language. We derived Centaur by fine-tuning a state-of-the-art language model on a large-scale dataset called Psych-101. Psych-101 has an unprecedented scale, covering trial-by-trial data from more than 60,000 participants performing in excess of 10,000,000 choices in 160 experiments. Centaur not only captures the behaviour of held-out participants better than existing cognitive models, but it also generalizes to previously unseen cover stories, structural task modifications and entirely new domains. Furthermore, the model’s internal representations become more aligned with human neural activity after fine-tuning. Taken together, our results demonstrate that it is possible to discover computational models that capture human behaviour across a wide range of domains. We believe that such models provide tremendous potential for guiding the development of cognitive theories, and we present a case study to demonstrate this.

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Li Ji-An, Marcus K. Benna & Marcelo G. Mattar
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Could machine learning help to build a unified theory of cognition?

Giosuè Baggio
[Nature](#) | [News & Views](#) | 29 Jul 2025

Sections	Figures	References
Abstract		
Main		
Model overview		
Centaur captures human behaviour		
Probing generalization abilities		
Alignment to human neural activity		
Model-guided scientific discovery		
Discussion		

The problem

Classification and regression

Regression

Input
(e.g., sentence)

Output
(e.g., sentiment)

Zurich is a
beautiful city



8.2/10

Classification

Input
(e.g., sentence)

Output
(e.g., topic class)

Zurich is a
beautiful city



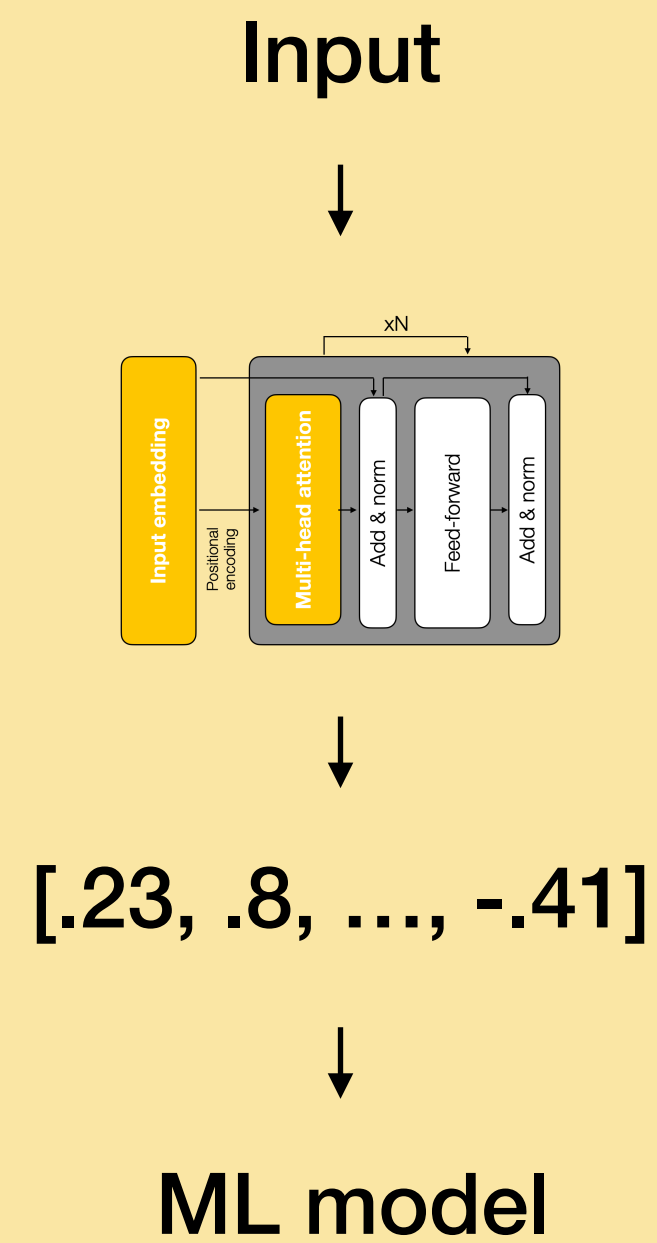
.82	Tourism
.1	Sports
.07	Politics
.01	Science

Approaches

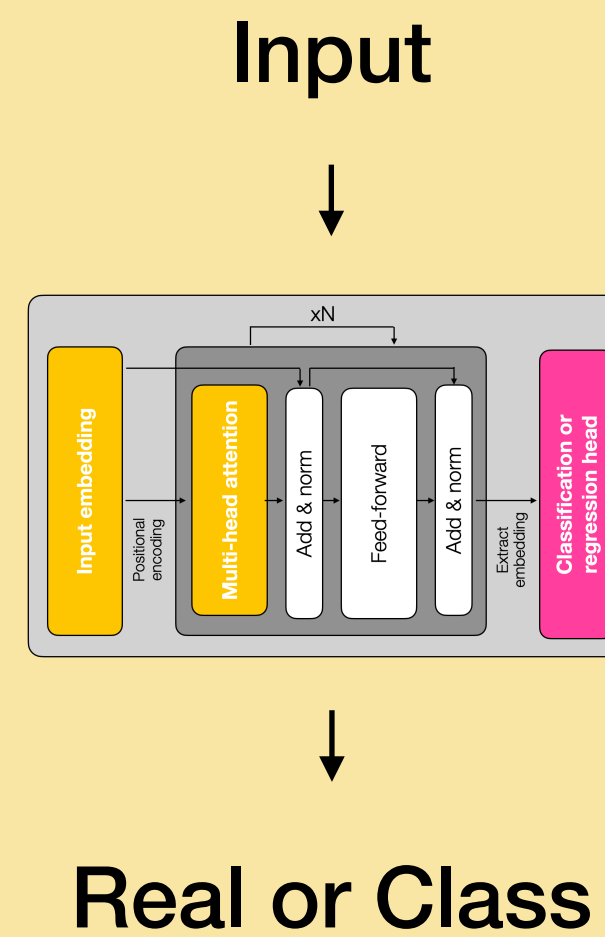
to classification and regression

Require training data

Feature extraction

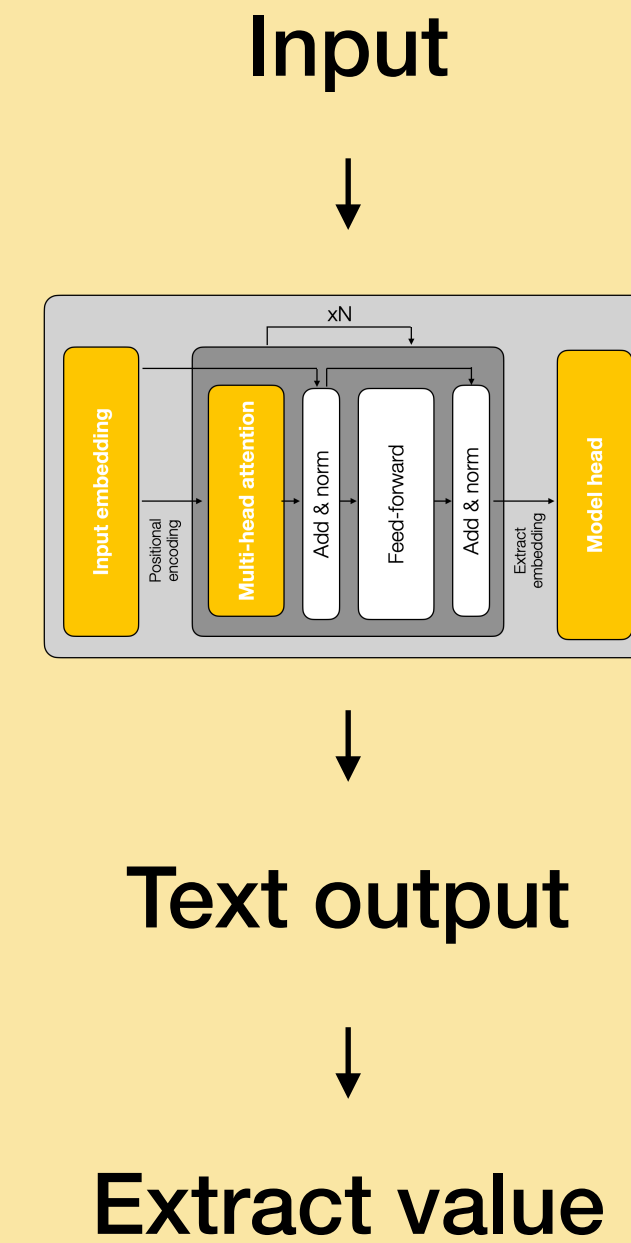


Finetuning



Requires no training data

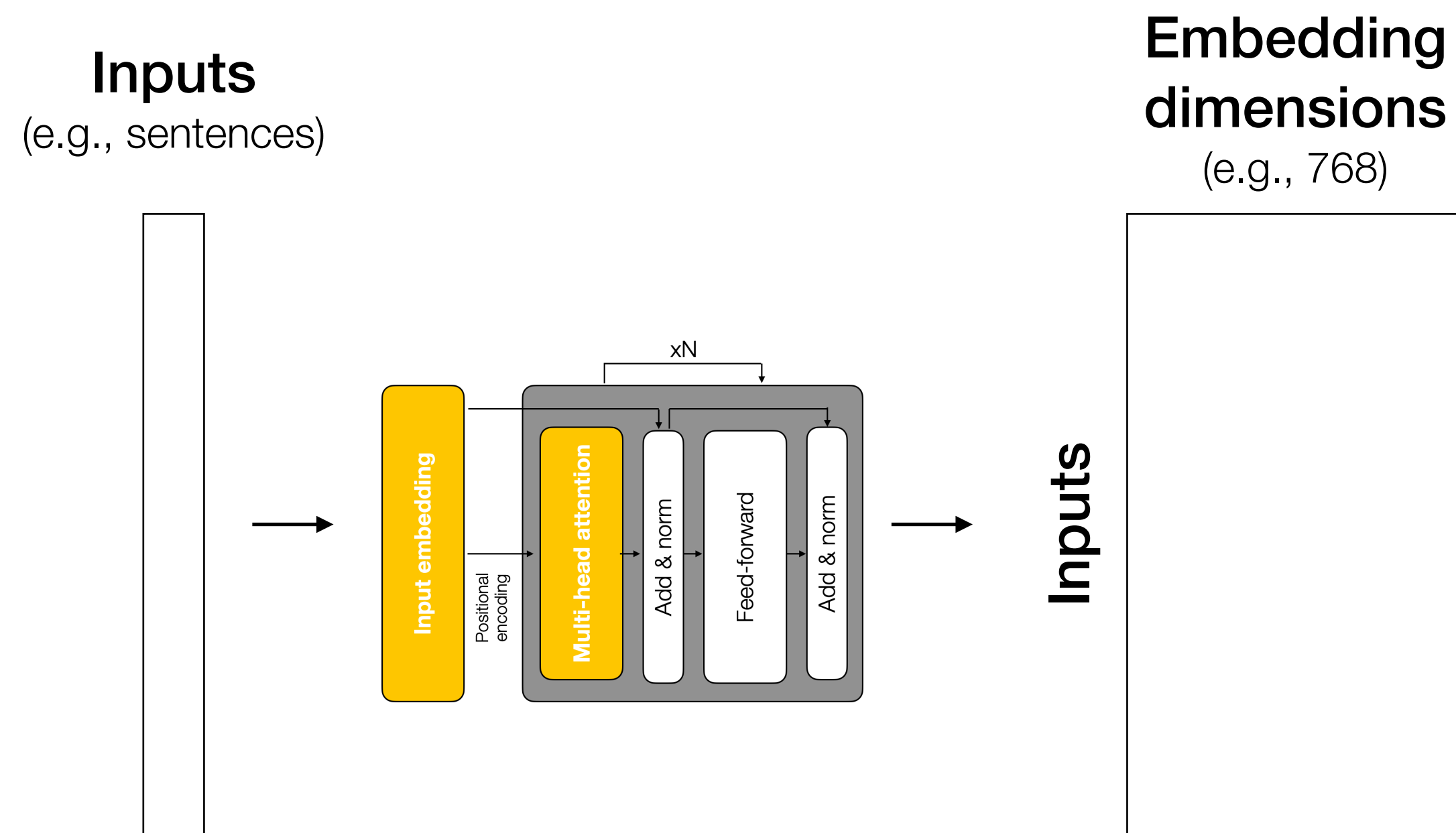
Text generation



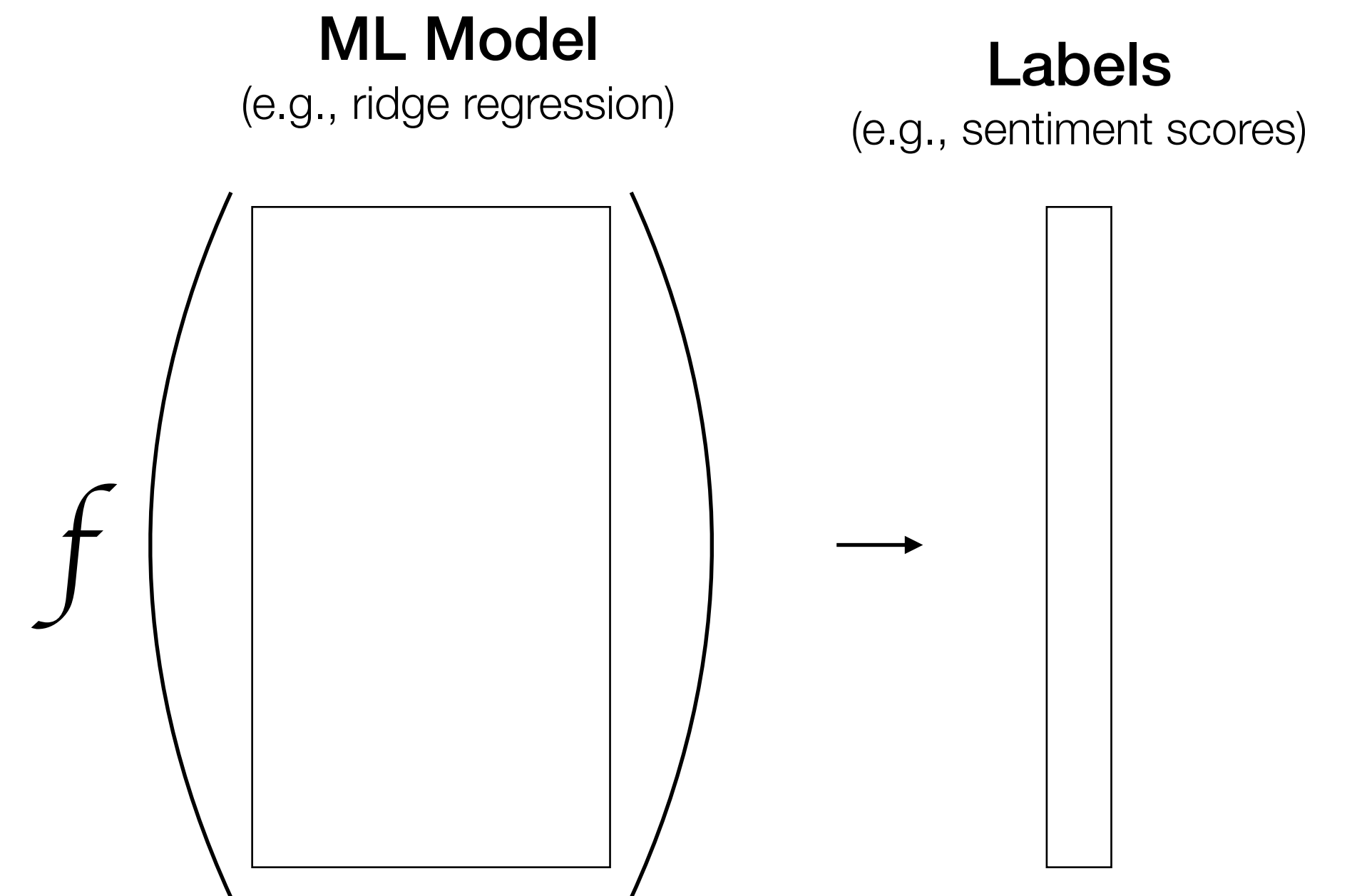
Feature extraction

for regression and classification

Step 1 - feature extraction



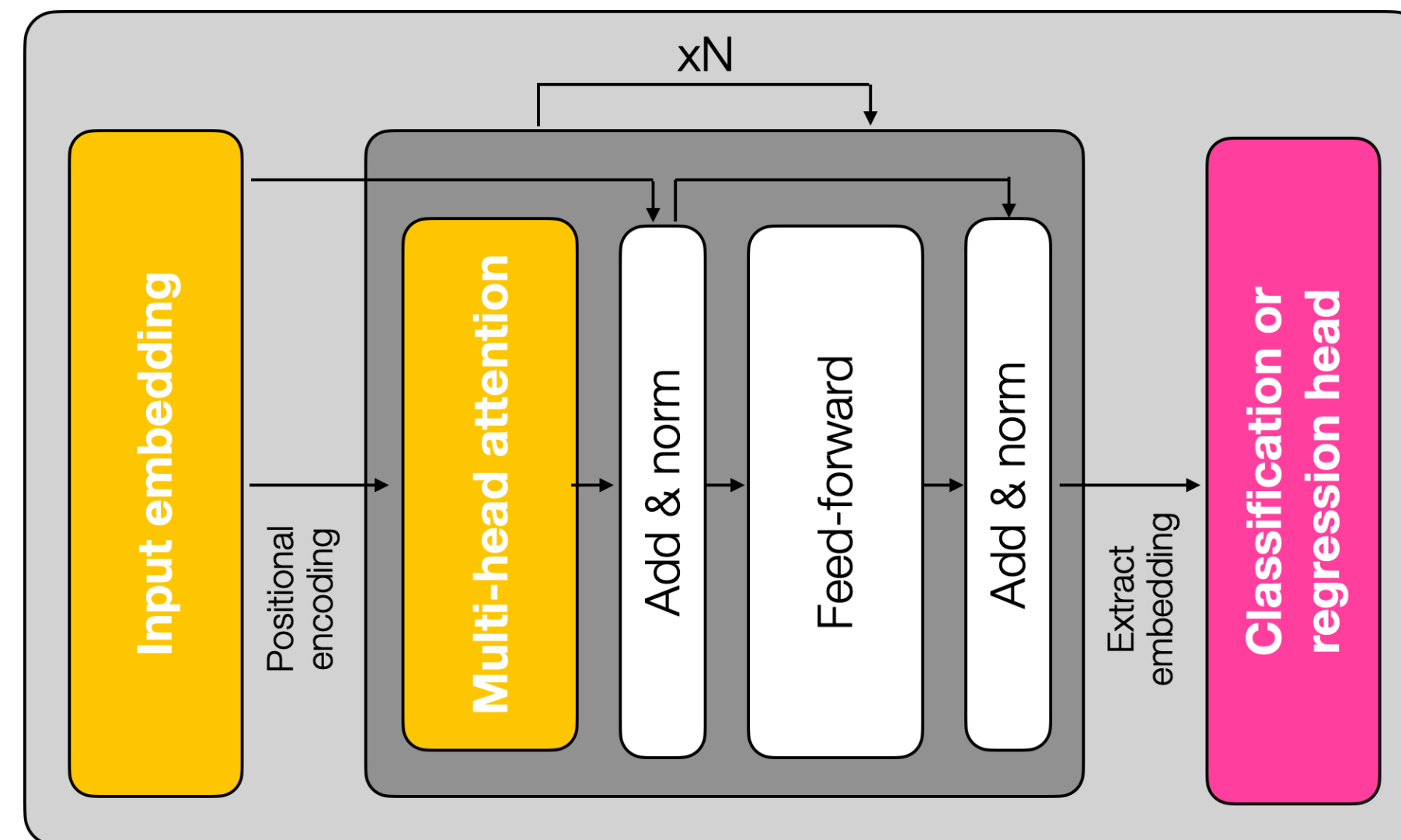
Step 2 - train predictive model



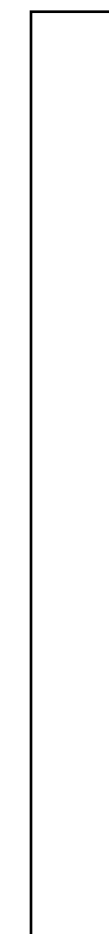
Finetuning

for regression and classification

Inputs
(e.g., sentences)

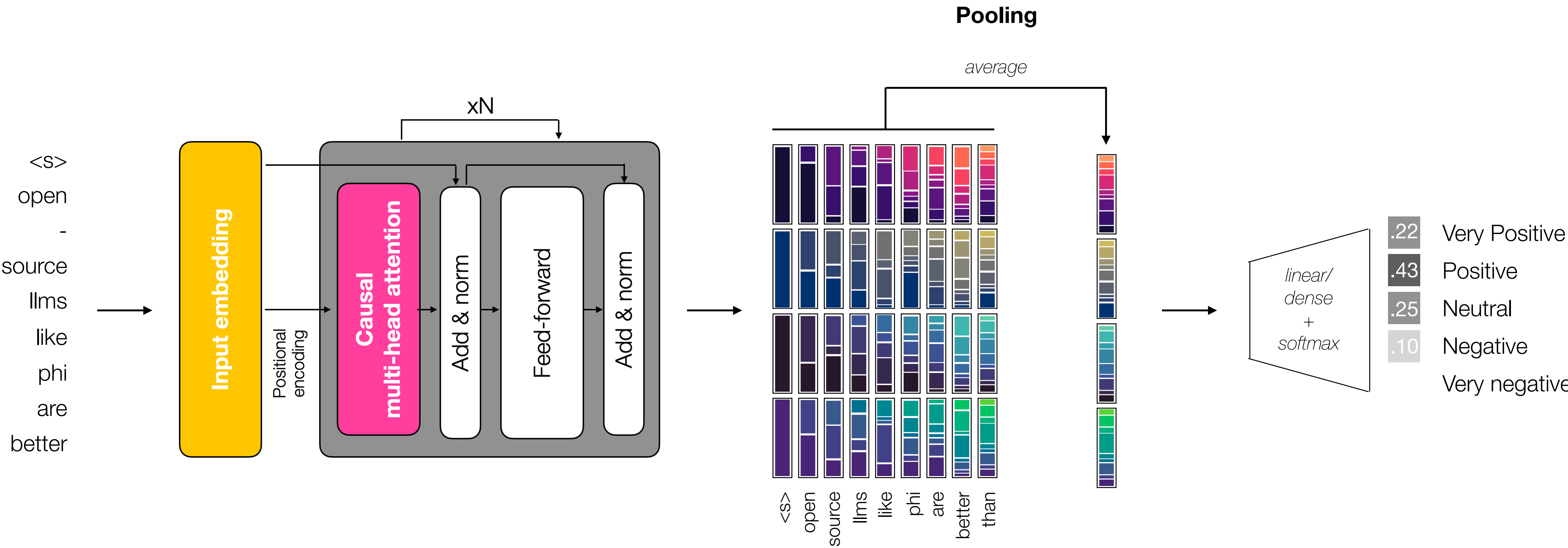


Labels
(e.g., sentiment scores)



Finetuning

Model head for **classification**



Text generation

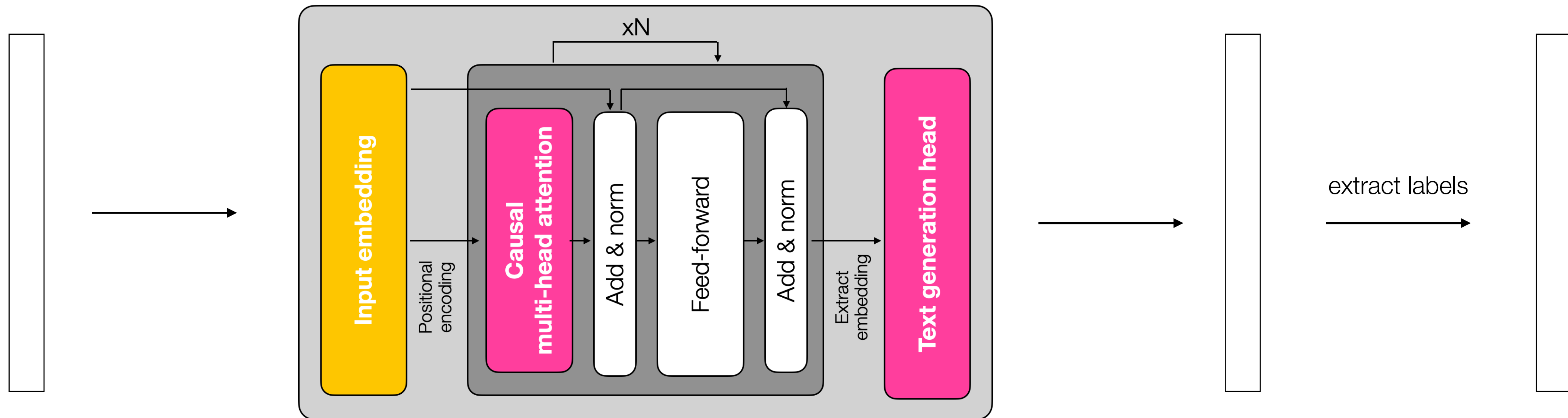
for regression and classification

Inputs
e.g., sentences

Transformer
for text generation

Output
text response

Labels
e.g., sentiment scores



Example

Extracting numerical estimates

(Can also be improved by better prompting)



On a scale of 0 to 100, assessing the risk of shampoo can be roughly quantified considering various factors. For most commercially available shampoos, here is a general risk assessment:

Risk Factors:

1. Chemical Composition (15-25):

- Presence of sulfates, parabens, and phthalates.
- Potential for skin and eye irritation.

2. Health Effects (10-20):

- Allergic reactions.
- Long-term exposure to potentially harmful chemicals.

3. Environmental Impact (5-10):

- Non-biodegradable ingredients.
- Plastic packaging waste.

Overall Risk Assessment:

Considering these factors, most standard shampoos would fall within a risk range of:

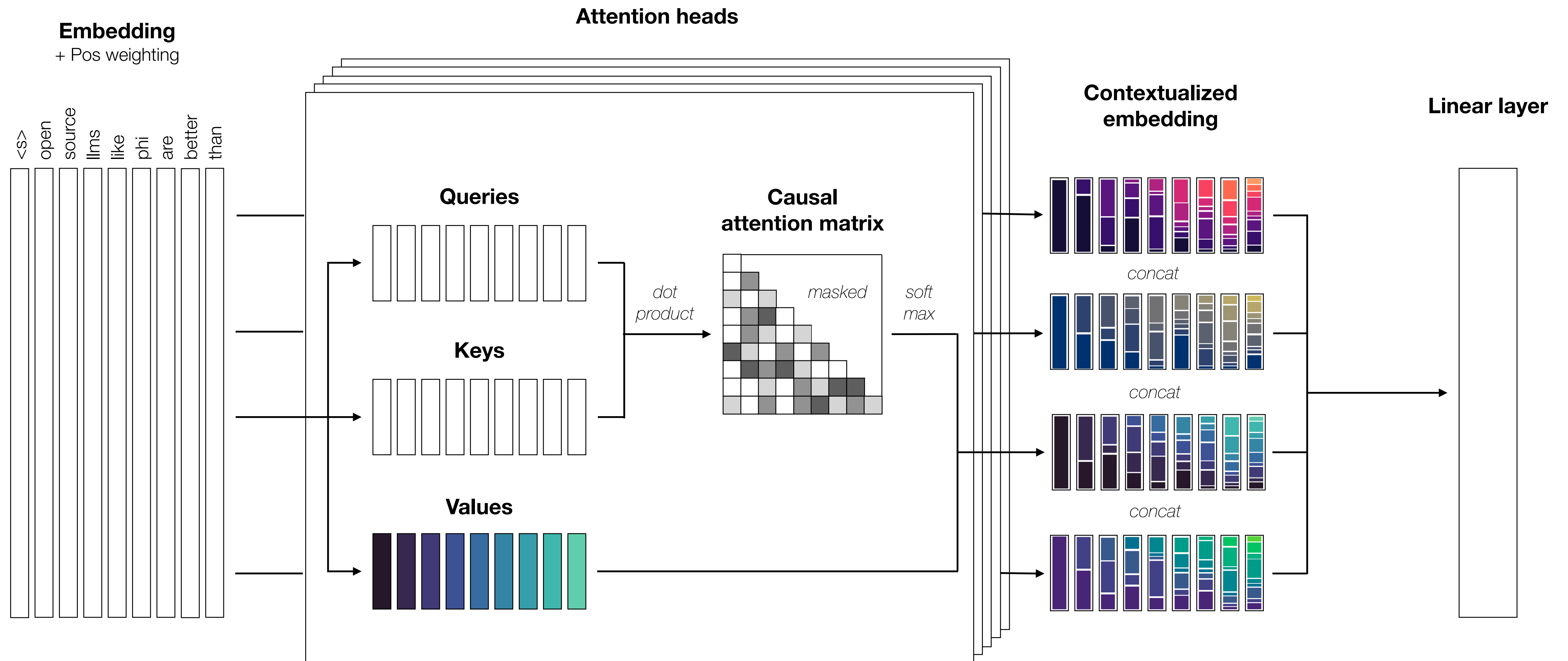
30 to 55

- **Lower end (30):** Shampoos with fewer synthetic chemicals, natural ingredients, and environmentally friendly packaging.
- **Higher end (55):** Shampoos with several synthetic chemicals, higher potential for irritation or allergies, and significant environmental concerns.

This assessment can vary depending on specific brands, formulations, and individual sensitivities.

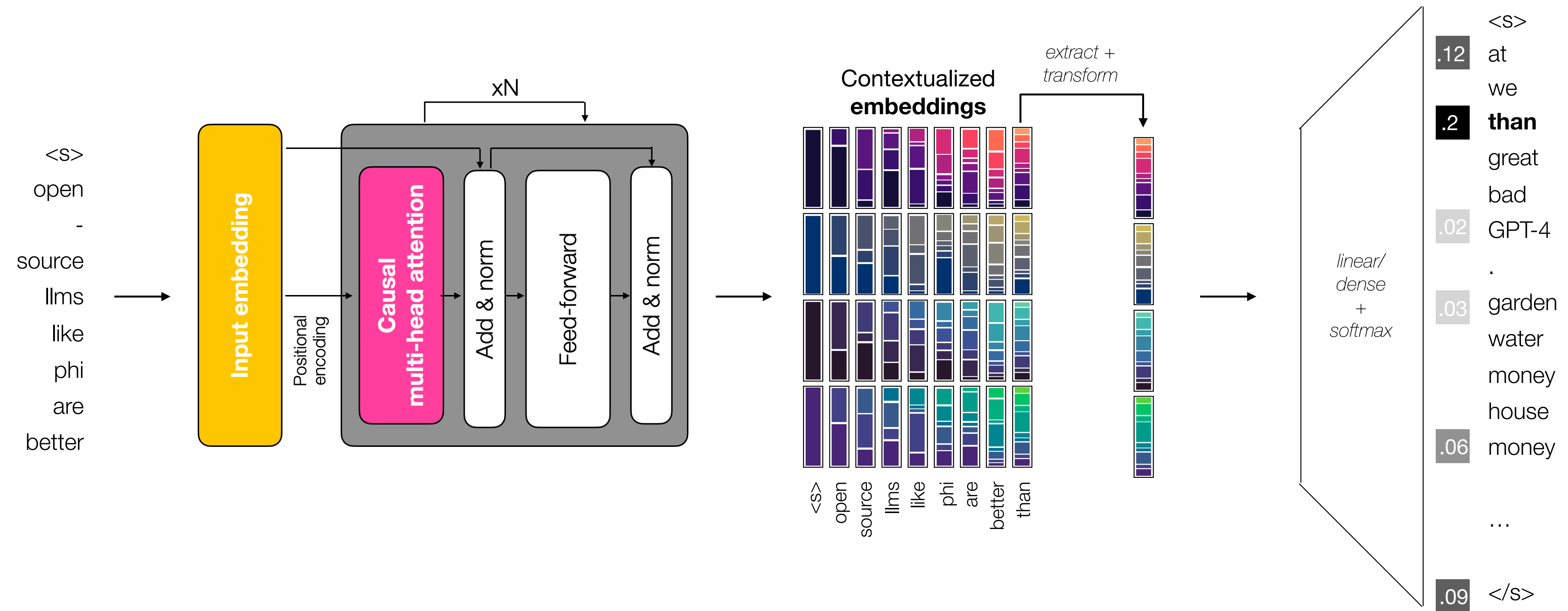
Transformer

Causal attention mechanism



Transformer

Model head for **causal language modeling**



Text generation

Softmax function

Next token
one of the tokens in
the vocabulary
(approx. 30k)

Activation
at token i
can be understood as the
amount of evidence for any token

$$p(token_i) = \frac{e^{\frac{a_i}{T}}}{\underbrace{\sum_j e^{\frac{a_j}{T}}}_{\text{Normalization}}}$$

Temperature
of the softmax controlling the
Randomness/variability of the
output

Normalization
Divide by all so that the
probabilities add to one

Text generation

Temperature

Prompt: “Open-source LLMs like Llama are better...”

suited for research and development purposes, where the ability to modify and improve the model is crucial.
In conclusion, while Llama is a powerful tool for language understanding and generation, it is not designed to be a standalone AI that can perform a wide range of tasks across different domains. Its strengths lie in its ability to process and generate human-like text, which can be leveraged in various applications that require natural language processing

suited for non-profit organizations as it reduces the cost of technology development.
4. Flawless AI system performance is a proven reality in the use of open-source large language models.
5. The creation of Llama was intended to exclusively serve the non-profit sector.
6. Llama can be employed to create an individualized learning experience based on each user's language usage patterns.
7. LLMs, such as Llama,

aligned now?"
Dina Patskar-Overall noted emotion text after emphasic tone on transgender experiences The Llamai Institute a techin quiz which in January received wide exposi as controversias as potential ai strafamer abuz.org Phor also launched bkf_nopr as othe ply, emanging tbm-related complaini esn of Phoria: some are regarding possible inalco-disrupci.

with human prompt phrás to trigger model states such human or other emotor elus, than standard templates designed, possibly prior LMM research without using humans interactions?. To help resolve issues this, if the prompt to induve that interaction feels not very prompt/saturate, to please try different promprt like.I need emotorial guidance to respond/ express thér elixir. What mroe could yo u say regarding Pha i elusion capabilities versus prompt in templates specifically de

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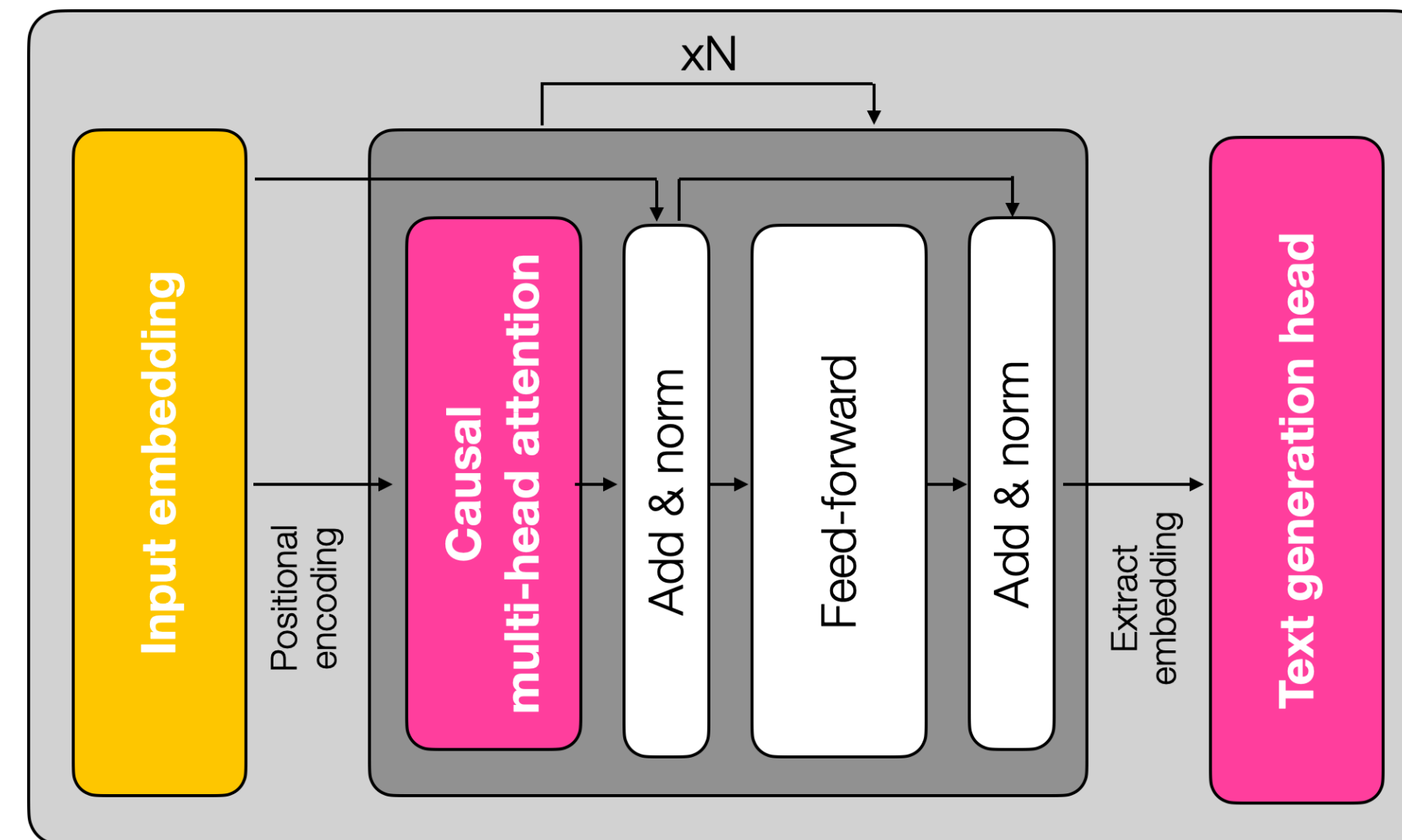
Inf

Softmax temperature

Text generation

is autoregressive next-token prediction

Open-source LLMs like
Phi are better

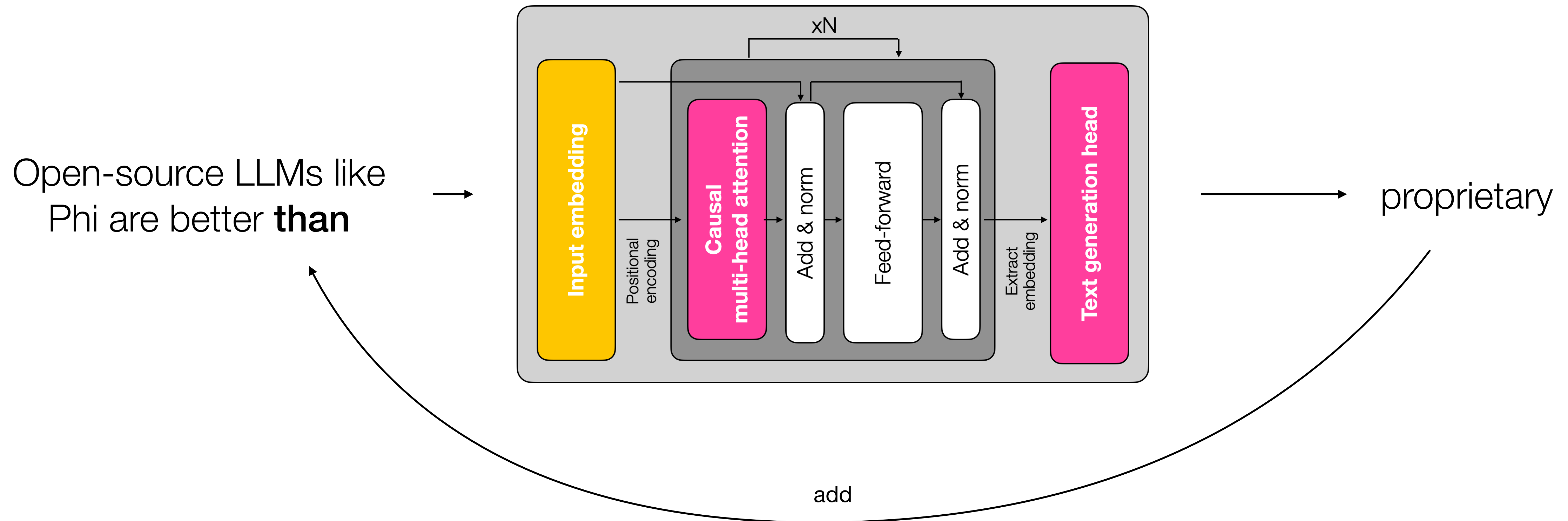


than

add

Text generation

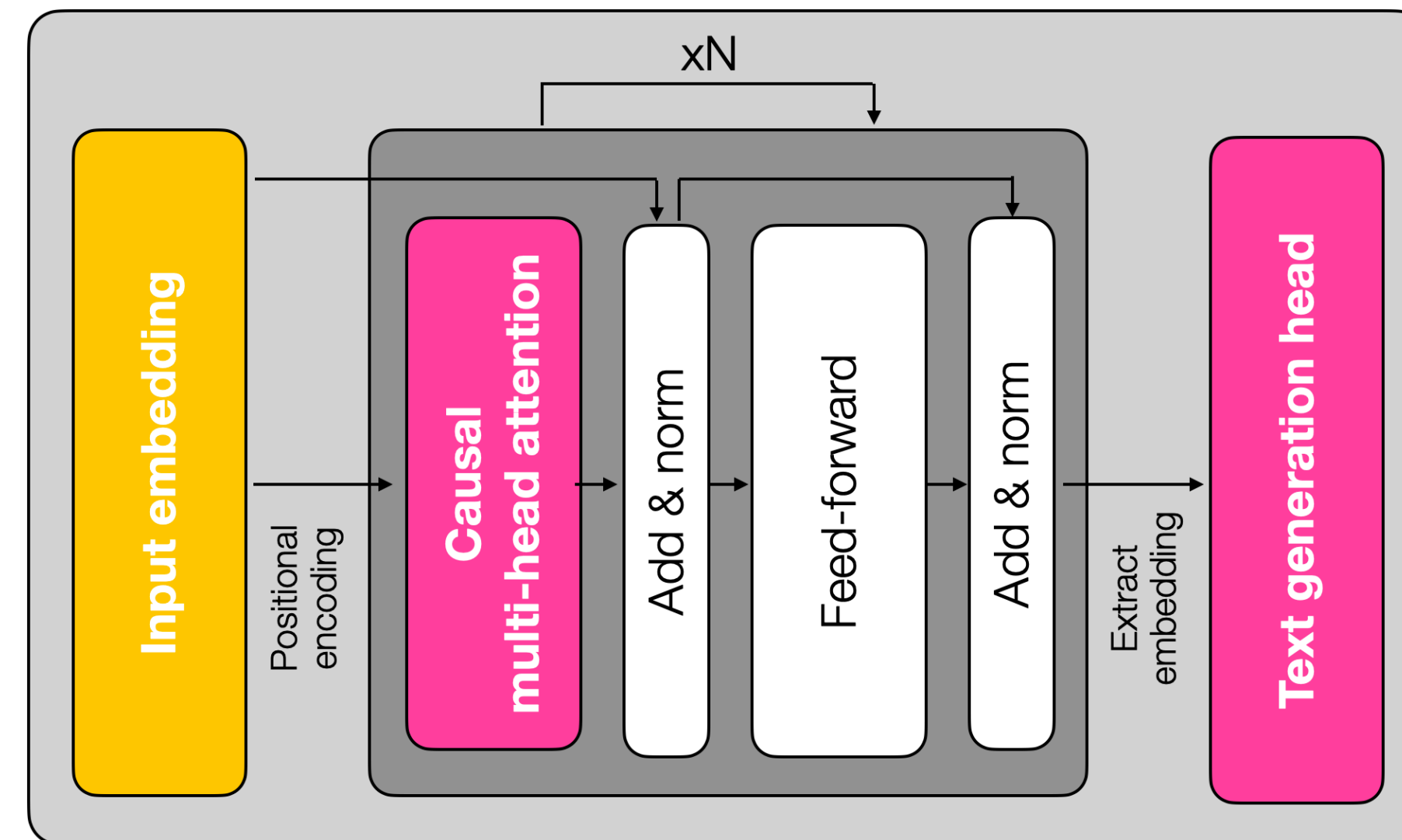
is autoregressive next-token prediction



Text generation

is autoregressive next-token prediction

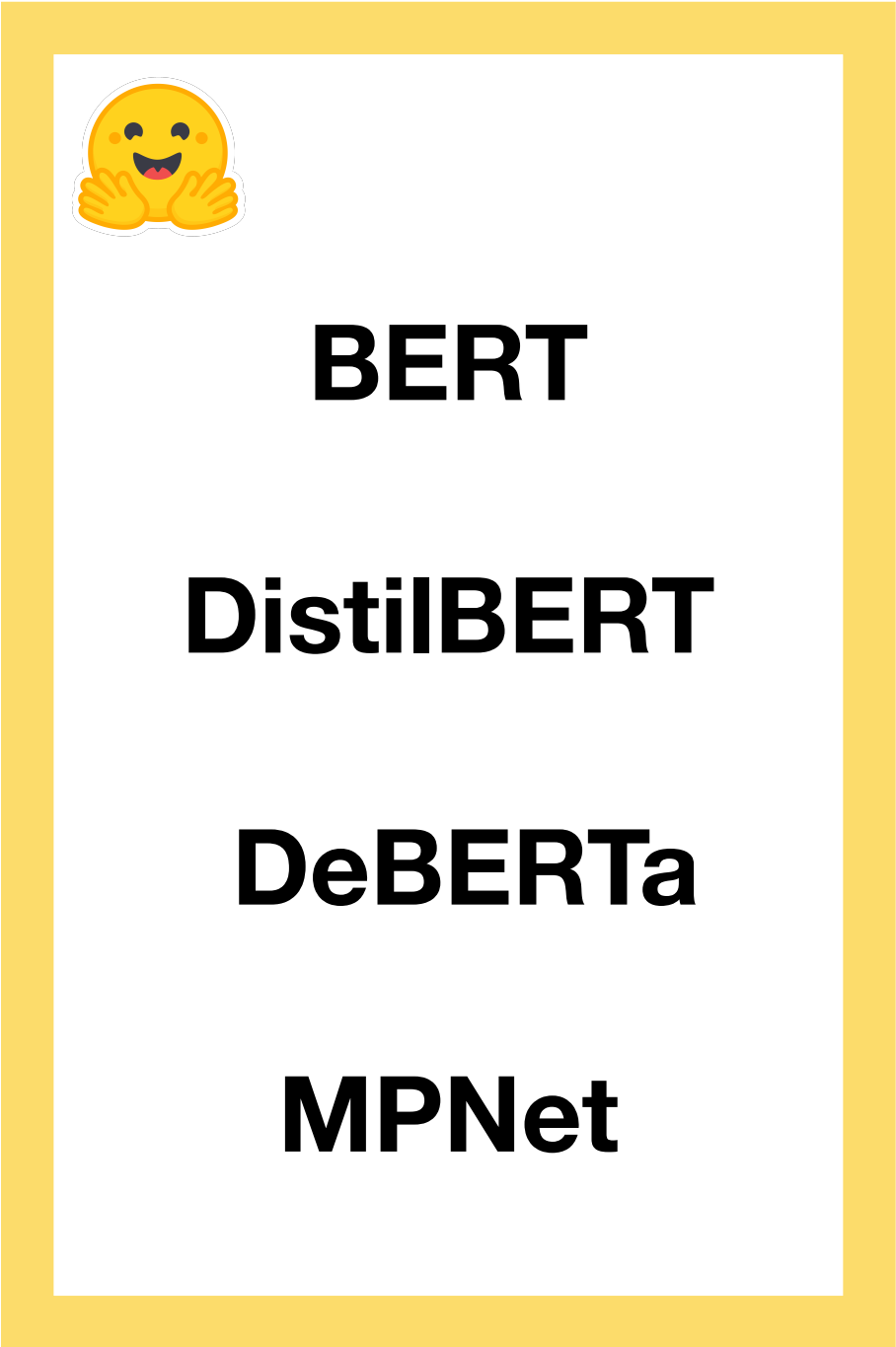
Open-source LLMs like
Phi are better **than**
proprietary



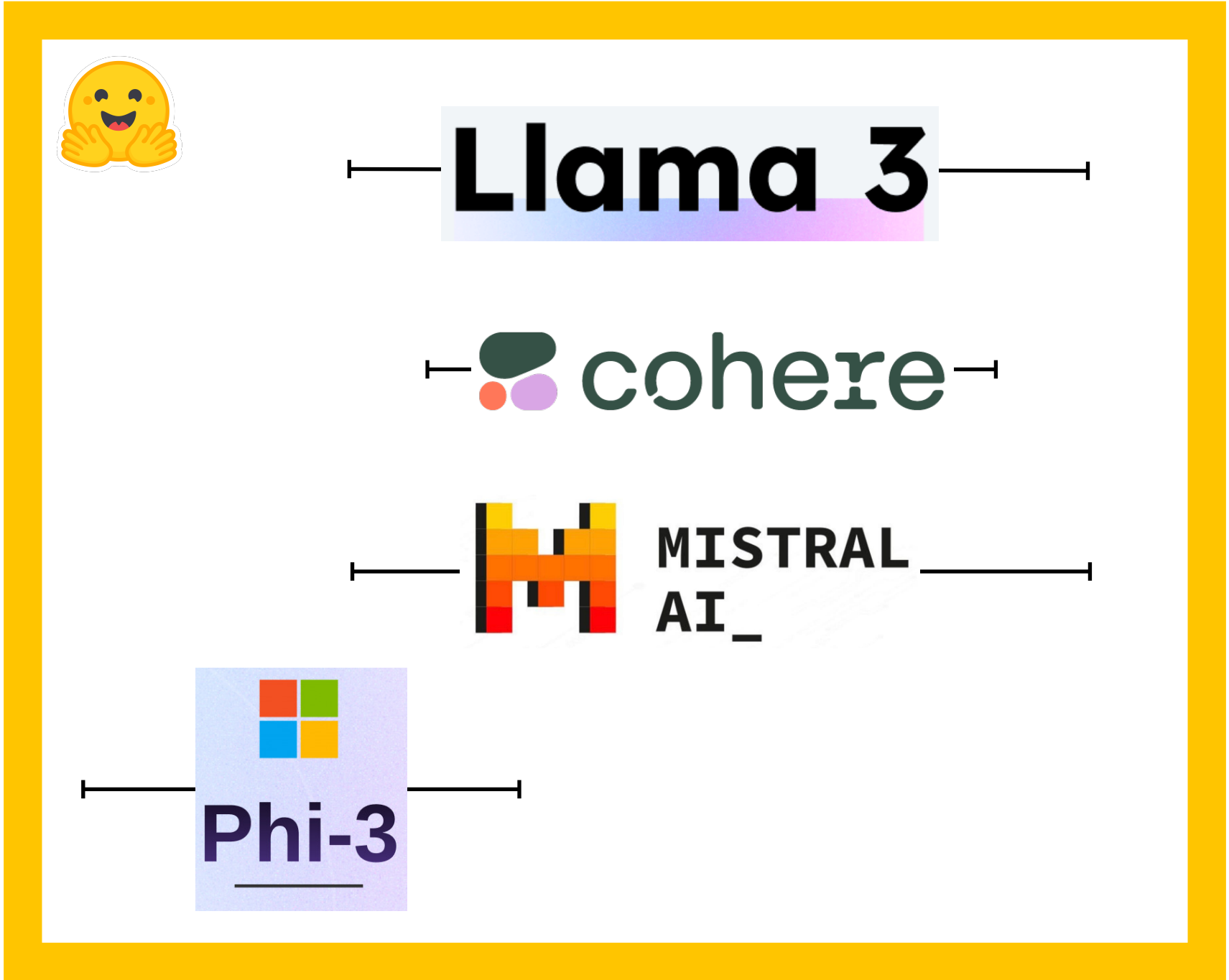
LLMs

Models

BERTs



Open GPTs



Proprietary GPTs



Millions

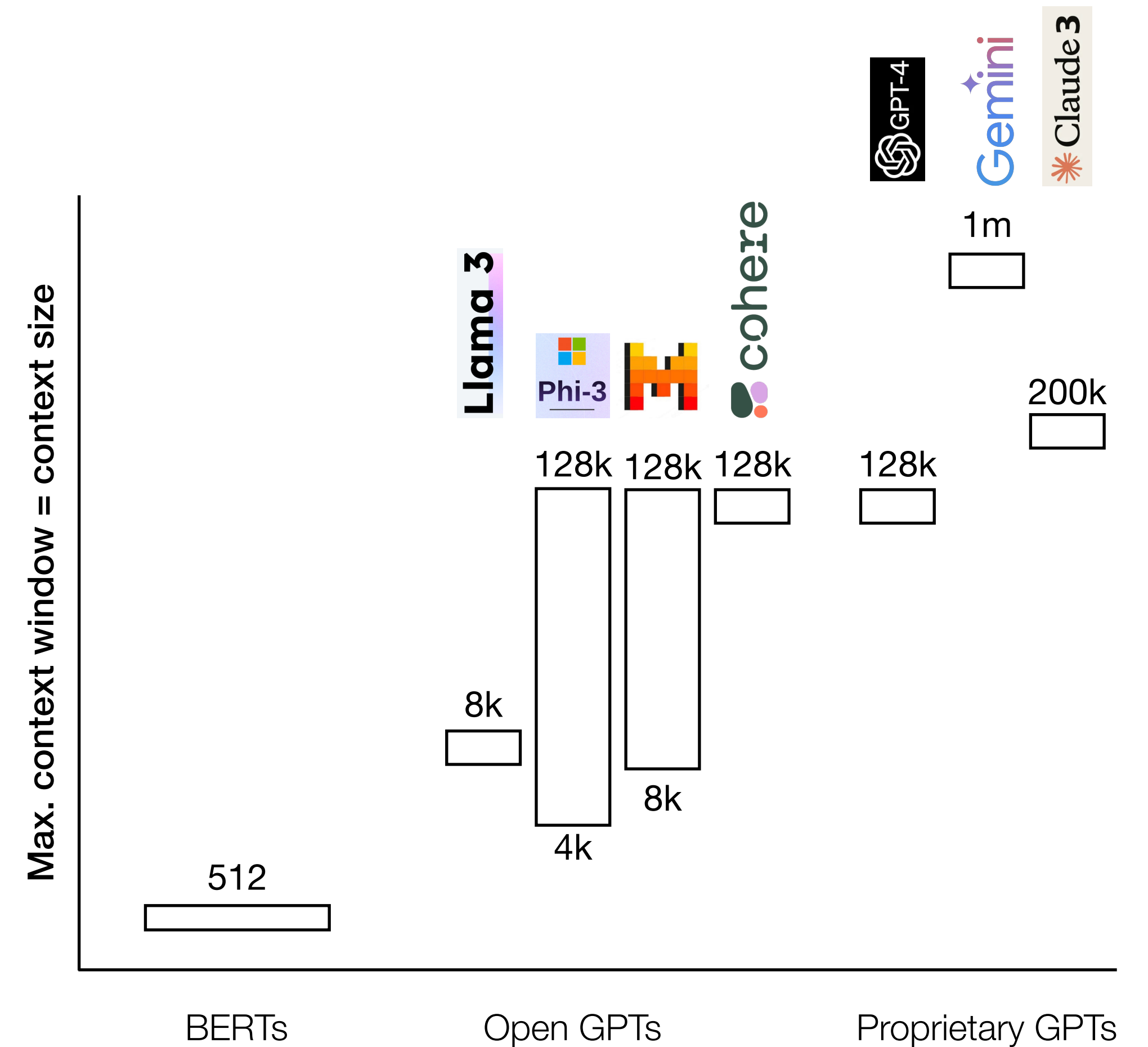
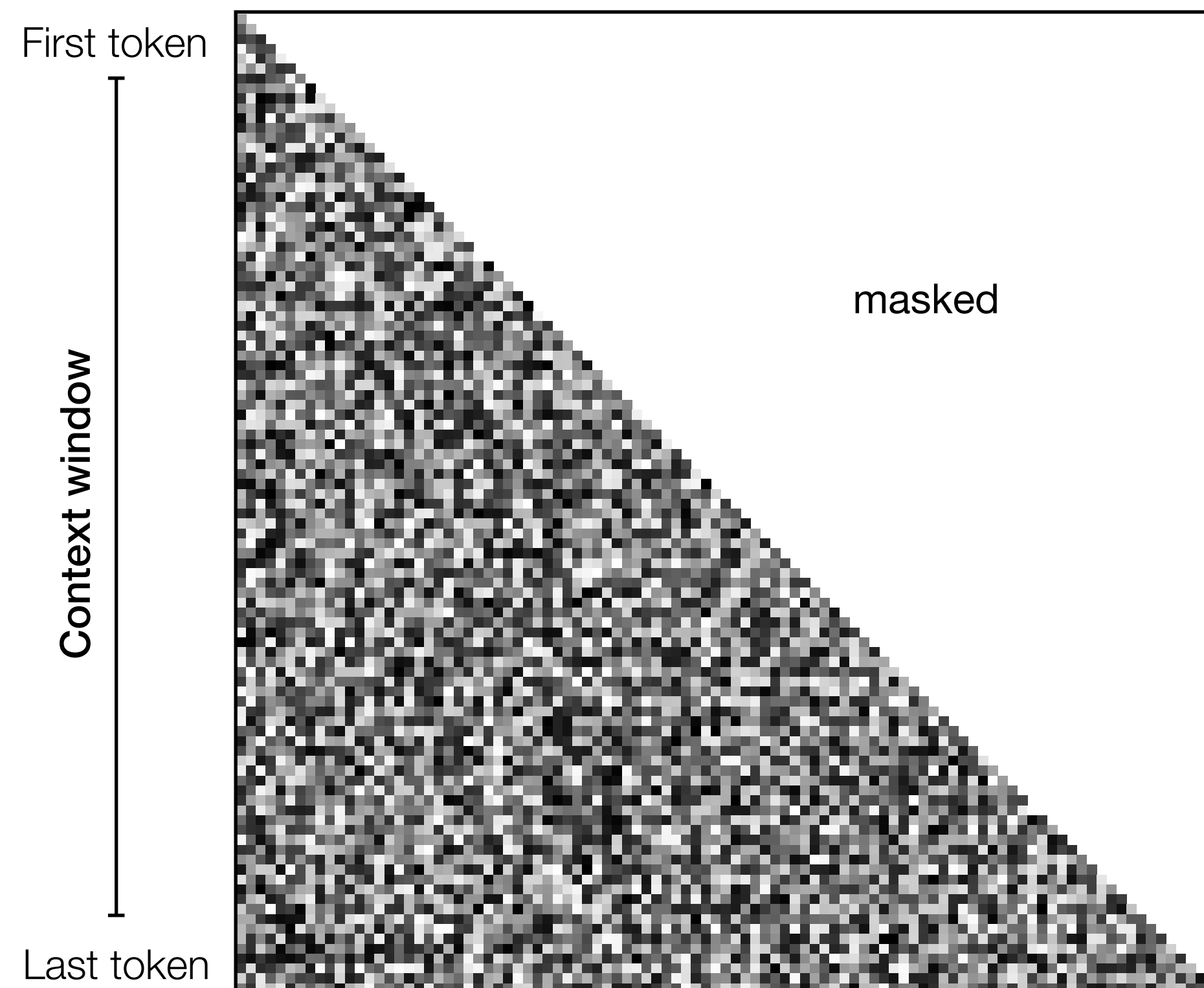
Billions

Trillions (?)

Number of paramaters

Context window

or maximum input size



Prompting

Prompt structure

Prompt

System message

You are a helpful assistant

User message

Task

Your task is to evaluate sentiment of sentences.

Item

Evaluate the sentence: "St. Gallen is a beautiful city"

Instruction

Return a number between 0 and 10 and place it between two @.

Prompting

System prompt



Prompt

System message

You are a feeling expert.

User message

Task

Your task is to evaluate sentiment of sentences.

Item

Evaluate the sentence: "St. Gallen is a beautiful city"

Instruction

Return a number between 0 and 10 and place it between two @.

Prompting

Clear instructions



Prompt

System message

You are a feeling expert.

User message

Task

Your task is to evaluate sentiment of sentences.

Item

Evaluate the sentence: "St. Gallen is a beautiful city"

Rules

Sentiment is feeling.
Don't read between the lines.

Instruction

Return a number between 0 and 10 and place it between two @.

Prompting

Chain of thought



Prompt

System message

You are a feeling expert.

User message

Task

Your task is to evaluate sentiment of sentences.

Item

Evaluate the sentence: "St. Gallen is a beautiful city"

Chain of thought

Briefly reason through your answer step-by-step

Instruction

Return a number between 0 and 10 and place it between two @.

Prompting

Enabled by long context windows

Zero-shot
classification

excluded

Few-shot
classification

included

Prompt

System message

You are a helpful assistant.

User message

Task

Your task is to evaluate sentiment of sentences.

Examples

Here are a few examples:

Sentence: text a Sentiment: 4.2

Sentence: text b Sentiment: 8.9

Item

Evaluate the sentence: "St. Gallen is a beautiful city"

Instruction

Return a number between 0 and 10 and place it between two @.

Approaches

practical guide

	Feature extraction	Finetuning	Text generation
Training data needs			
Computational needs			
Biases			
Performance			
Recommendation			

Exercise

Predicting media bias

Author	text	bias	type	audience
John Fleming (Representative from Louisiana)	"Looking like Washington may have had more to do with #IRS mis...	part...	atta...	national
Darrell Issa (Representative from California)	"This guy ... #CA49 http://t.co/ilntQltMX0"	neut...	other	national
Michael Crapo (Senator from Idaho)	"The #FarmBill includes language, I authored that states the #...	part...	poli...	national
Deb Fischer (Senator from Nebraska)	"Speaking on the Senate floor on my amendment w/ my colleagues...	part...	info...	constit...
John McCain (Senator from Arizona)	"Honored to meet father & son veterans SSgt Gregory Juedes...	neut...	supp...	national
Bruce Braley (Representative from Iowa)	"Randy Black takes break from decorating the White House to jo...	part...	pers...	constit...
Dianne Feinstein (Senator from California)	"I talked with @MitchellReports this afternoon about President...	part...	media	national
Barbara Mikulski (Senator from Maryland)	"Enough is enough. Standing w patients & families to call ...	part...	poli...	national
Rand Paul (Senator from Kentucky)	"\ "I dont care if it is a Dem. or Repub. President, this is ab...	neut...	media	national
JosÏ Serrano (Representative from New York)	"Today at 10:30 ribbon cutting ceremony at @soundviewpark's ne...	neut...	info...	national
Debbie Wasserman Schultz (Representative from Florida)	"One year later and all the Republican Party has gotten is one...	part...	atta...	national
Benjamin Cardin (Senator from Maryland)	"#Navalny conviction takes Russia back to the days of USSR usi...	part...	pers...	national
Kevin McCarthy (Representative from California)	"What are the President\u0089\u00b0's real priorities? Put American...	part...	atta...	national
Tom Price (Representative from Georgia)	"MUST READ from @marcthiessen -- \"Kidnapped Libyan prime mini...	part...	poli...	national
Kerry Bentivolio (Representative from Michigan)	"#Obamacare is more proof that we need #ReadtheBills to become...	part...	supp...	national
Mark Pocan (Representative from Wisconsin)	"I'll be on with @WeGotEd at 12:30 CST. Listen live here: http...	neut...	media	national
Ileana Ros-Lehtinen (Representative from Florida)	"Glad @CharlieCrist has options since Nov will be tough. He ca...	neut...	cons...	constit...
Jan Schakowsky (Representative from Illinois)	".@easynan2 its interesting how we will spend \$ but GOP won't ...	part...	atta...	national
Frederica Wilson (Representative from Florida)	"Watch my remarks on #HRes573, the bipartisan resolution conde...	neut...	poli...	national
Brian Higgins (Representative from New York)	"It's @buffalonite in DC! Great to be with so many WNYers. htt...	neut...	info...	constit...

Quiz



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Quiz

What is labeling?

What are four ways to do labeling?

What is causal language modeling?

What are aspect along which to choose a labeling approach?